

Comparing the growth kinetics and characteristics of Wharton's jelly derived mesenchymal stem cells expanded using different culture mediums.

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Wharton's jelly-derived mesenchymal stem cells (WJ-MSCs) can be isolated from umbilical cords which is abundant and easy to obtain. Due to their potent immunosuppressive properties, multilineage differentiation potential, and lack of ethical issues, WJ-MSCs are considered a promising candidate for therapeutic applications. However, large-scale in vitro expansion is necessary to obtain enough cells for therapeutic purposes. Therefore, this study aimed to optimize cell culture conditions and determine the characteristics of expanded WJ-MSCs. WJ-MSCs were seeded in 6-well plates at a density of 5000 cells/cm² and cultured with different mediums, including DMEM-LG+10% FBS, DMEM-LG+10% HPL, serum-free commercial medium 1, serum-free GMP grade commercial medium 2, and HPL supplemented commercial medium 3. The cell morphology and growth kinetics were compared, and the three most suitable mediums were selected for further experiments. WJ-MSCs were then cultured in the selected mediums at seeding densities ranging from 1000 to 5000 cells/cm², and cell growth kinetics were analysed. WJ-MSCs cultured in the selected mediums were characterized by their immunophenotype, tri-lineage differentiation potential and immunosuppression property. WJ-MSCs cultured with DMEM-LG+10% HPL, commercial medium 1 and commercial medium 2 showed smaller size, significantly higher cell yield, and shorter population doubling time than those cultured in other mediums. Hence, these three mediums were selected for further experiments. Only DMEM-LG + 10% HPL medium maintained high cell yields ($1.48 \pm 0.14 \times 10^6$ with bFGF and $1.56 \pm 0.17 \times 10^6$ without bFGF) at the lowest seeding density tested. However, WJ-MSCs cultured in all three mediums expressed the MSC surface markers, were able to suppress PBMC proliferation, and could differentiate into adipogenic, chondrogenic and osteogenic lineages. In summary, DMEM-LG+10% HPL is the best medium for WJ-MSC expansion, as it provides the highest cell yield without compromising cell characteristics and functionality. The potential of this medium for large-scale expansion using a bioreactor or multilayered flask should be investigated in the future.

Human umbilical cord mesenchymal stem cells: a new era for stem cell therapy.

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The human umbilical cord is a promising source of mesenchymal stem cells (HUCMSCs). Unlike bone marrow stem cells, HUCMSCs have a painless collection procedure and faster self-renewal properties. Different derivation protocols may provide different amounts and populations of stem cells. Stem cell populations have also been reported in other compartments of the umbilical cord, such as the cord lining, perivascular tissue, and Wharton's jelly. HUCMSCs are noncontroversial sources compared to embryonic stem cells. They can differentiate into the three germ layers that promote tissue repair and modulate immune responses and anticancer properties. Thus, they are attractive autologous or allogenic agents for the treatment of malignant and nonmalignant solid and soft cancers. HUCMCs also can be the feeder layer for embryonic stem cells or other pluripotent stem cells. Regarding their therapeutic value, storage banking system and protocols should be established immediately. This review critically evaluates their therapeutic value, challenges, and future directions for their clinical applications.

Stem cells from umbilical cord and placenta for musculoskeletal tissue engineering.

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Mesenchymal stem cells isolated from amnion/amniotic fluid, umbilical cord blood, placental tissue, umbilical cord vein and the Wharton's Jelly are promising candidates for musculoskeletal tissue engineering of bone and cartilage tissues. The extracorporeal nature of this source avoids the ethical concerns that plague the isolation of embryonic stem cells. Moreover, the harvesting does not require the invasive and discomfort extraction procedures as well as patient risks that attend adult stem cell isolation. Current preclinical studies support the application of these cell-based therapies for the regeneration of musculoskeletal tissues. We performed a review of the literature to focus on actual knowledge and the future perspectives of the stem cells deriving from umbilical cord and placenta for musculoskeletal tissue engineering.

Wharton's jelly-derived stromal cells and their cell therapy applications in allogeneic haematopoietic stem cell transplantation.

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For decades, mesenchymal stromal cells (MSCs) have been of great interest in the fields of regenerative medicine, tissue engineering and immunomodulation. Their tremendous potential makes it desirable to cryopreserve and bank MSCs to increase their accessibility and availability. Postnatally derived MSCs seem to be of particular interest because they are harvested after delivery without ethical controversy, they have the capacity to expand at a higher rate than adult-derived MSCs, in which expansion decreases with ageing, and they have demonstrated immunological and haematological supportive properties similar to those of adult-derived MSCs. In this review, we focus on MSCs obtained from Wharton's jelly (the mucous connective tissue of the umbilical cord between the amniotic epithelium and the umbilical vessels). Wharton's jelly MSCs (WJ-MSCs) are a good candidate for cellular therapy in haematology, with accumulating data supporting their potential to sustain haematopoietic stem cell engraftment and to modulate alloreactivity such as Graft Versus Host Disease (GVHD). We first present an overview of their in-vitro properties and the results of preclinical murine models confirming the suitability of WJ-MSCs for cellular therapy in haematology. Next, we focus on clinical trials and discuss tolerance, efficacy and infusion protocols reported in haematology for GVHD and engraftment.

Wharton's jelly-derived mesenchymal cells as a new source for the generation of microtissues for tissue engineering applications.

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Microtissues (MT) are currently considered as a promising alternative for the fabrication of natural, 3D biomimetic functional units for the construction of bio-artificial substitutes by tissue engineering (TE). The aim of this study was to evaluate the possibility of generating mesenchymal cell-based MT using human umbilical cord Wharton's jelly stromal cells (WJSC-MT). MT were generated using agarose microchips and evaluated ex vivo during 28 days. Fibroblasts MT (FIB-MT) were used as control. Morphometry, cell viability and metabolism, MT-formation process and ECM synthesis were assessed by phase-contrast microscopy, functional biochemical assays, and histological analyses. Morphometry revealed a

time-course compaction process in both MT, but WJSC-MT resulted to be larger than FIB-MT in all days analyzed. Cell viability and functionality evaluation demonstrated that both MT were composed by viable and metabolically active cells, especially the WJSC during 4-21 days ex vivo. Histology showed that WJSC acquired a peripheral pattern and synthesized an extracellular matrix-rich core over the time, what differed from the homogeneous pattern observed in FIB-MT. This study demonstrates the possibility of using WJSC to create MT containing viable and functional cells and abundant extracellular matrix. We hypothesize that WJSC-MT could be a promising alternative in TE protocols. However, future cell differentiation and in vivo studies are still needed to demonstrate the potential usefulness of WJSC-MT in regenerative medicine.

Mesenchymal Stem Cells from Wharton's Jelly and Amniotic Fluid.

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The discovery of mesenchymal stem cells (MSCs) in perinatal sources, such as the amniotic fluid (AF) and the umbilical connective tissue, the so-called Wharton's jelly (WJ), has transformed them into promising stem cell grafts for the application in regenerative medicine. The advantages of AF-MSCs and WJ-MSCs over adult MSCs, such as bone marrow-derived mesenchymal stem cells (BM-MSCs), include their minimally invasive isolation procedure, their more primitive cell character without being tumourigenic, their low immunogenicity and their potential autologous application in congenital disorders and when cryopreserved in adulthood. This chapter gives an overview of the biology of AF-MSCs and WJ-MSCs, and their regenerative potential based on the results of recent preclinical and clinical studies. In the end, open questions concerning the use of WJ-MSCs and AF-MSCs in regenerative medicine will be emphasized.

Comparative Analysis of Porcine Adipose- and Wharton's Jelly-Derived Mesenchymal Stem Cells.

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Mesenchymal stem cells (MSCs) are promising candidates for tissue regeneration, cell therapy, and cultured meat research owing to their ability to differentiate into various lineages including adipocytes, chondrocytes, and osteocytes. As MSCs display different characteristics depending on the tissue of origin, the appropriate cells need to be selected according to the purpose of the research. However, little is known of the unique properties of MSCs in pigs. In this study, we compared two types of porcine mesenchymal stem cells (MSCs) isolated from the dorsal subcutaneous adipose tissue (adipose-derived stem cells (ADSCs)) and Wharton's jelly of the umbilical cord (Wharton's jelly-derived mesenchymal stem cells (WJ-MSCs)) of 1-day-old piglets. The ADSCs displayed a higher proliferation rate and more efficient differentiation potential into adipogenic and chondrogenic lineages than that of WJ-MSCs; conversely, WJ-MSCs showed superior differentiation capacity towards osteogenic lineages. In early passages, ADSCs displayed higher proliferation rates and mitochondrial energy metabolism (measured based on the oxygen consumption rate) compared with that of WJ-MSCs, although these distinctions diminished in late passages. This study broadens our understanding of porcine MSCs and provides insights into their potential applications in animal clinics and cultured meat science.

A simple and serum-free protocol for cryopreservation of human umbilical cord as source of Wharton's jelly mesenchymal stem cells.

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Mesenchymal stromal cells (MSCs) show promise in cell-based transplantations and regenerative medicine applications. MSCs from Wharton's jelly (WJ) of umbilical cord can be easily harvested and exhibit greater proliferative activity than bone marrow MSCs. It is important to develop a practical cryopreservation technique to effectively store umbilical cord for potential future applications. Successful cryopreservation would allow access to umbilical cord from the same donor for repeated WJ MSC-based transplantations. For therapeutic applications, one should be able to obtain clinically-relevant quality and quantity of MSCs from cryopreserved tissues. In this study, we optimised a serum-free formulation of 10% dimethyl sulfoxide (DMSO) and 0.2M sucrose for cryopreservation of umbilical cord tissue. Slow freezing and rapid thawing were adopted. MSCs harvested from WJ of cryopreserved umbilical cord could undergo robust expansion, differentiate to mesodermal lineages and express MSC-characteristic surface antigens. The cumulative cell yield, however, was less compared to corresponding fresh cord tissue.

Mesenchymal stem cells in the Wharton's jelly of the human umbilical cord.

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The Wharton's jelly of the umbilical cord contains mucoid connective tissue and fibroblast-like cells. Using flow cytometric analysis, we found that mesenchymal cells isolated from the umbilical cord express matrix receptors (CD44, CD105) and integrin markers (CD29, CD51) but not hematopoietic lineage markers (CD34, CD45). Interestingly, these cells also express significant amounts of mesenchymal stem cell markers (SH2, SH3). We therefore investigated the potential of these cells to differentiate into cardiomyocytes by treating them with 5-azacytidine or by culturing them in cardiomyocyte-conditioned medium and found that both sets of conditions resulted in the expression of cardiomyocyte markers, namely N-cadherin and cardiac troponin I. We also showed that these cells have multilineage potential and that, under suitable culture conditions, are able to differentiate into cells of the adipogenic and osteogenic lineages. These findings may have a significant impact on studies of early human cardiac differentiation, functional genomics, pharmacological testing, cell therapy, and tissue engineering by helping to eliminate worrying ethical and technical issues.

Isolation and Culture of Three Kinds of Umbilical Cord Mesenchymal Stem Cells.

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Umbilical cord mesenchymal stem cells (UC-MSCs) are an important cell source for regenerative medicine. UC-MSCs can be isolated from the umbilical cord Wharton's jelly, as well as from the umbilical arteries and umbilical vein. They are known as perivascular stem cells obtained from umbilical arteries (UCA-PSCs), perivascular stem cells obtained from the umbilical vein (UCV-PSCs), and mesenchymal stem cells obtained from Wharton's jelly (WJ-MSCs). UCA-PSCs and UCV-PSCs are pericytes derived from perivascular regions that are progenitors of MSCs. Isolation and culture of the three kinds of cells is an important source for studying stem cell transplantation and repair. The present protocol focuses on the isolation and culture of cells through mechanical separation, adherent culture, and cell crawling out.

Through this technique, the three different types of stem cells can be derived. Cell surface markers were detected by flow cytometry. The stem cells were detected for multilineage differentiation potential by adipogenic, osteogenic, and neural-like differentiation, which is consistent with the phenotype of MSCs. This experimental protocol expands the source of UC-MSCs. In addition, the cell isolation method provides a basis for further study of regenerative medicine and other applications.
